Lab Report

Lab 1.7

a. The problem during this question was examining the 2012 NFL season, whether or not a kick was successful for that season. The solution using a Linear Regression model and inserting all the data as a CSV file, creating input and output parameters and predicting via input if the kick was successful or not.

b. For my code, it initially says to create a dataframe for the X (inputs) so I had created the dataframe X using basic syntax, similarly creating a dataframe y
Then initializing the model of LinearRegression() then we fit it into the model after we flatten the y dataframe. Then using the scikit learn we printed out the output for the prediction and the score of the model.

Lab 2.6

a. In this problem we are dealing with the Sloan Digital Sky Survey, the dataset consist of 10,000 Objects and we have two features which are redshift and u_g
b. For the code I have to import a few things from the scikit learn library as well as pandas and Numpy,
Then create the dataset X and Y which are [“redshift”, “u_g”] and [“classes”].
We initialize the KNeighboursClassifer defining n_neighbours as 3 according to the question.

1. Confusion Matrix: lets take the user input seed as 41

   [[1449 7 28]
   [10 235 0]
   [2 0 1269]]

2. Macro F1 Score: 0.979

3. (**1,449**, **235**, and **1,269** correct predictions). This indicates that the 3-NN classifier successfully identified the vast majority of deep sky objects based solely on their redshift and engineered u_g color profile.

   Macro F1 Score of 0.979 is extremely high. Because a macro average calculates the metric for each class independently and averages them with equal weight

Lab 2.7

a. This lab addresses a binary classification task to determine whether a basketball team will win or lose an NBA match. Utilizing a massive historical dataset containing **126,314 games** played between 1947 and 2015
b. First we import all the necessary libraries from scikit learn as well as numpy and pandas.
   Then we have to load the nbaallelo_log.csv into the dataframe then create binary feature for game result with 0 and 1
   The initialize the model with a penalty of 12 according to the question, then we fit the model with relevant columns
   The target feature is processed via np.ravel(y) inside the .fit() call. This collapses the pandas column into a flat, 1-D contiguous array to fulfill scikit-learn's input matrix requirements.
   Then at the end we accessing .coef_ and .intercept_ exposes the fitted coefficients to solve the probability sigmoid curve
c. [[42413 20744]

   [20744 42413]]

d. Binary F1 Score (Win Class): 0.671

e. This symmetry in the confusion matrix reflects that the historical dataset contains an exact 50/50 split of wins and losses, every game must have one winner and one loser
   The F1 score of 0.671 indicates that logistic classifier is a solid baseline model which performes better than a 50-50 guess on who’s going to win